SkillHub • Master Project Documentation & Viva Voce Guide

Enterprise Full-Stack Learning Management System: Architectural Walkthrough, Technical Defense, Evaluation Rubrics & Viva Defense Cheat Sheet

ASP.NET Core 8 Web API | Angular 20 Standalone | Clean Architecture | Hybrid RBAC + PBAC | Google Gemini AI | Viva & Presentation Ready

SkillHub – Enterprise Learning & Course Management Platform

Master Project Documentation, Architecture Guide & Viva Cheatsheet

1. Executive Summary & Project Vision

SkillHub is an enterprise-grade, full-stack digital learning management platform designed and built using **ASP.NET Core 8 Web API** and **Angular 20 (Standalone Architecture)**. The platform provides an end-to-end ecosystem catering to five distinct personas: **Students**, **Instructors**, **SubAdmins**, **SuperAdmins**, and **Product Support Specialists**.

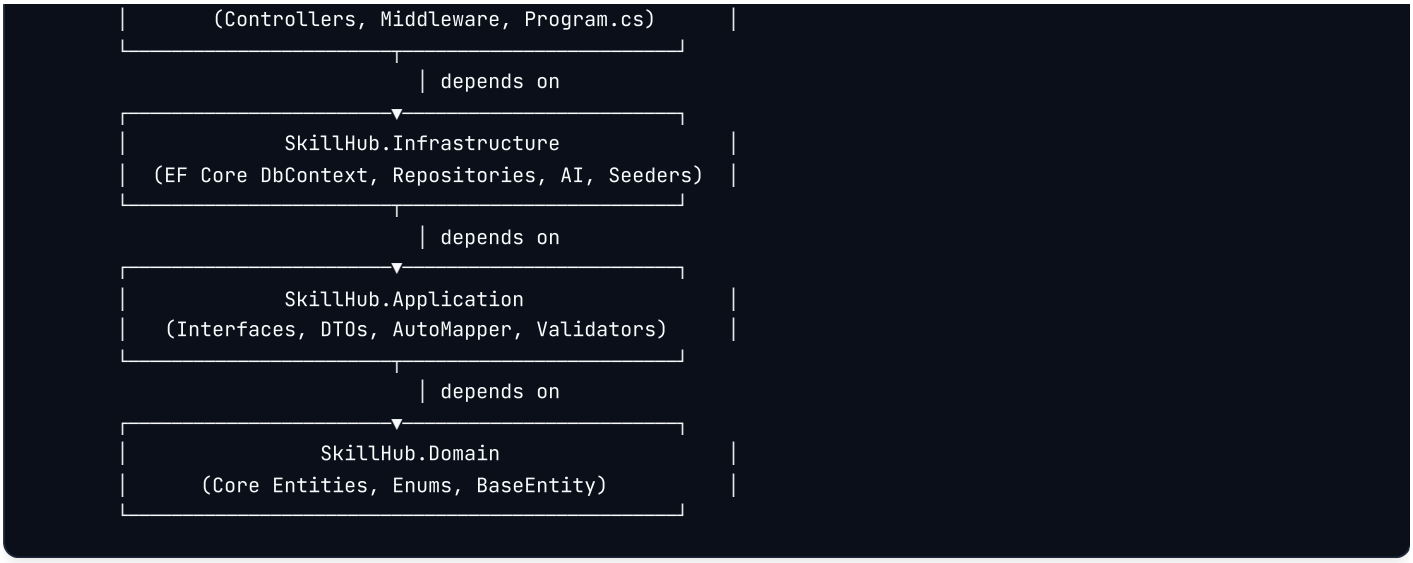
Core Platform Capabilities

- **Curriculum & Course Authoring**: Instructors build multi-section courses with streaming video lessons, downloadable PDF resources, and structured syllabi.
- **Interactive Learning Player**: Students stream video lectures, toggle completion status, track progress in real time, and download reference materials.
- **Mastery Assessments & 70% Rule Certification**: Automated quiz engine with multiple-choice questions. Certificates are locked until the student finishes **100% of lessons** and achieves **≥ 70% score** on all assessments.
- **Enterprise Helpdesk & Support Triage**: Role-governed ticketing system where **Product Support** exclusively triages tickets, assigns them to Admins, or solves them directly, while **SuperAdmins** have strict read-only oversight.
- **Permission Studio (RBAC + PBAC)**: Dynamic permission management where platform privileges can be inspected and updated without recompiling or redeploying code.
- **Google Gemini AI Learning Copilot**: Integrated AI tutor providing contextual learning guidance, smart prompt suggestions, and student assistance with guest quota controls.
- **Integrated Payment Checkout**: Bank-grade checkout gateway supporting Cards, UPI QR Code, and Net Banking with instant atomic enrollment unlocking.

2. System Architecture & Clean Architecture Deep-Dive

SkillHub adheres strictly to **Clean Architecture** (Onion / Hexagonal Architecture principles). The core tenet is the **Dependency Inversion Principle**: high-level business logic does not depend on low-level implementation details (such as database engines or UI frameworks); rather, both depend on abstractions.





Why Clean Architecture Over Traditional 3-Tier Architecture?

1. **Separation of Concerns**: Business rules in `SkillHub.Application` and `SkillHub.Domain` are independent of SQL Server, Entity Framework, or ASP.NET Core controllers.
2. **Testability**: Core logic can be unit-tested without mocking a live database or spinning up web servers.
3. **Maintainability & Evolution**: If SQL Server is replaced with PostgreSQL, or if Angular is replaced with a mobile app (Flutter/MAUI), the Domain and Application layers remain untouched.

Layer Breakdown in the Codebase

1. `SkillHub.Domain` (Core Business Objects)

- **Zero dependencies** on external libraries or frameworks.
- Contains enterprise business entities, value objects, and domain enums.
- **Key Files**:
- `BaseEntity.cs`: Common audit fields (`Id`, `CreatedAt`, `UpdatedAt`, `CreatedBy`, `UpdatedBy`).
- `Entities/Identity/`: `User.cs`, `Role.cs`, `UserRole.cs`, `Permission.cs`, `RolePermission.cs`, `UserPermission.cs`, `RefreshToken.cs`.
- `Entities/Courses/`: `Course.cs`, `CourseSection.cs`, `CourseLesson.cs`, `LessonResource.cs`, `CourseCategory.cs`, `Enrollment.cs`.
- `Entities/Learning/`: `LessonCompletion.cs`, `CourseReview.cs`, `Certificate.cs`.
- `Entities/Assessments/`: `Quiz.cs`, `QuizQuestion.cs`, `QuizOption.cs`, `QuizAttempt.cs`, `QuizAnswer.cs`.
- `Entities/Support/`: `SupportTicket.cs`, `TicketMessage.cs`.

2. `SkillHub.Application` (Business Orchestration Contracts)

- Defines the use cases, service contracts (interfaces), transfer objects (DTOs), mappings, and validation rules.
- **Key Files**:
- `Features/Courses/DTOs/`: `CourseDtos.cs` (DTOs for creating, updating, and viewing courses).
- `Features/Learning/DTOs/`: `LearningDtos.cs` (DTOs for player progress, lesson completion, certificates).
- `Features/Assessments/DTOs/`: `QuizDtos.cs` (DTOs for quiz questions, submission, scoring).
- `Features/Support/DTOs/`: `SupportDtos.cs` (DTOs for ticketing, status updates, assignment).
- `Features/Ai/DTOs/`: `AiChatDtos.cs` (DTOs for Gemini requests, responses, suggestions).

- `Mappings/MappingProfile.cs`: AutoMapper configuration mapping Domain Entities to DTOs.
- `Features/*/Validators/`: FluentValidation classes (`CreateCourseValidator.cs`, `LoginRequestValidator.cs`, etc.).

3. `SkillHub.Infrastructure` (Data Access, External Services & Persistence)

- Implements the contracts defined in the Application layer.
- Handles EF Core data access, database migrations, database seeding, password hashing, and third-party integrations (Google Gemini, SMTP).
- **Key Files**:
- `Persistence/Context/ApplicationDbContext.cs`: EF Core database context configuring all 19 DbSets and relationship constraints.
- `Services/Learning/LearningService.cs`: Orchestrates lesson toggles, completion percentages, and certificate evaluation.
- `Services/Assessments/QuizService.cs`: Manages quiz questions, calculates attempt scores, and records student attempts.
- `Services/Support/SupportService.cs`: Handles ticket creation, staff messaging, status workflows, and admin assignment.
- `Services/Ai/GeminiAiService.cs`: Google Gemini REST API client with streaming prompt engineering and guest quota caching.
- `Services/Admin/PermissionService.cs`: Provides catalog data for Permission Studio and handles role-permission bindings.
- `seed/`: `RoleSeeder.cs`, `UserSeeder.cs`, `PermissionSeeder.cs`, `CategorySeeder.cs`.

4. `SkillHub.API` (Presentation & HTTP Delivery)

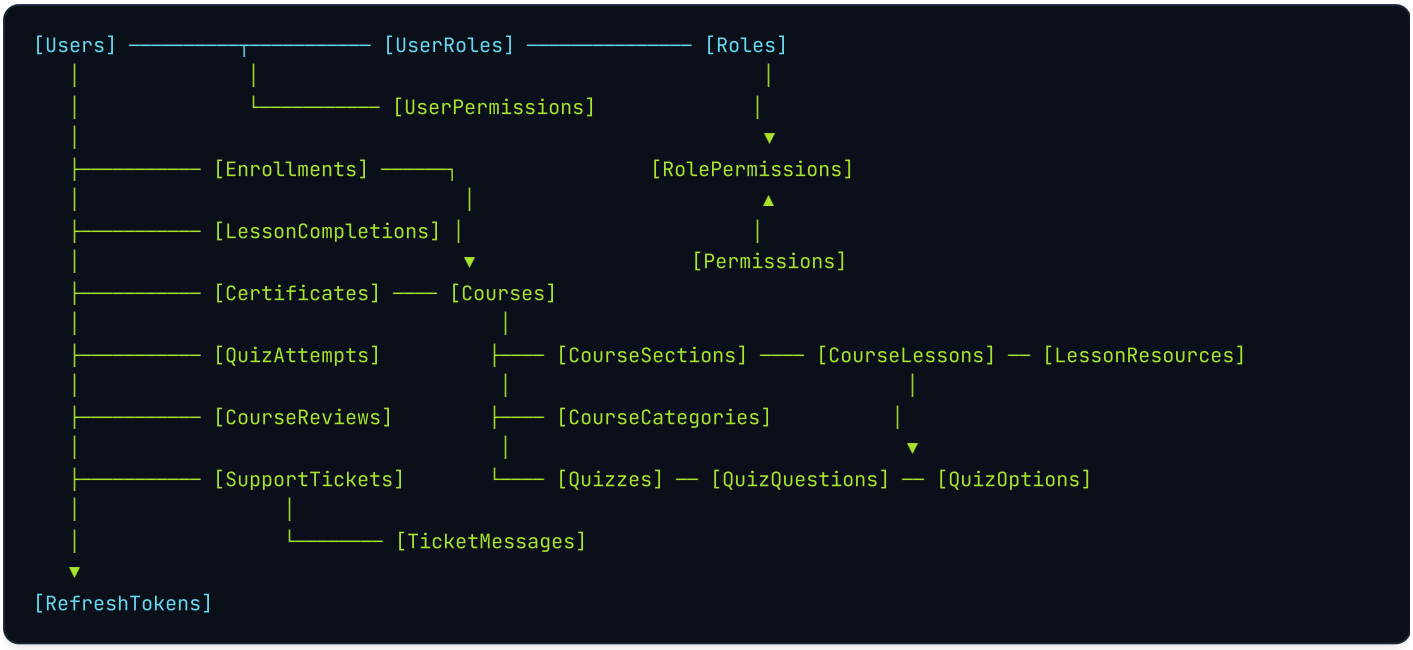
- Exposes RESTful endpoints, handles HTTP authentication headers, cross-origin resource sharing (CORS), and global exception handling.
- **Key Files**:
- `Controllers/AuthController.cs`: Registration, login, refresh tokens, password reset.
- `Controllers/CourseController.cs`: Course browsing, filtering, search, and authoring.
- `Controllers/LearningController.cs`: Enrolled player data, lesson completion toggles.
- `Controllers/QuizzesController.cs`: Quiz retrieval, answer submission, assessment scores.
- `Controllers/SupportController.cs`: Ticket creation, queue viewing, status transitions, admin assignment.
- `Controllers/AdminController.cs`: User governance, role assignment, course approvals.
- `Controllers/PermissionController.cs`: Permission Studio endpoints.
- `Controllers/AiChatController.cs`: Gemini AI interaction endpoint.
- `Program.cs`: ASP.NET Core startup, dependency injection container setup, JWT configuration, CORS policy.

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3. Database Architecture & What Makes It Unique

The database uses **Entity Framework Core 8 Code-First** with Microsoft SQL Server. It is engineered with strict relational integrity, defensive delete constraints, and audit capabilities.

Entity Relationship Diagram (ERD Summary)



What Makes Our Database Architecture Distinct?

UNIQUE ARCHITECTURE FEATURE	HOW SKILLHUB IMPLEMENTS IT	REAL-WORLD BENEFIT
Defensive Delete Constraints (`OnDelete(DeleteBehavior.Restrict)`)	Configured in `ApplicationDbContext.cs` for Users, Tickets, and Assigned Admins.	Prevents cascading deletes from accidentally wiping student learning history or ticket audit logs if a user is deleted.
Hybrid RBAC + PBAC	Normalized `RolePermissions` and `UserPermissions` join tables.	Enables both role grouping (e.g. `ProductSupport`) and individual user permission overrides (e.g. granting a specific SubAdmin approval power).
Strict Assessment Auditing	`QuizAttempts` and `QuizAnswers` tables record exact historical attempts and timestamps.	Allows students to review their mistakes while preventing cheating or score manipulation.
Deterministic Unique Ticket Numbers	Unique indexed ticket numbers (`SKILLHUB-TKT-XXXX`) generated with random cryptographic suffixes.	Clean, human-readable identifiers for students and support agents.
Automated Seeding Pipeline	System seeds default roles, admin credentials, standard categories, and 30+ granular permissions on startup.	Zero-setup onboarding: the platform can spin up on a fresh database with fully working demo data.

4. Role Governance & Access Control Matrix

SkillHub implements a role model tailored to enterprise operations:

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          SKILLHUB ROLE GOVERNANCE                        │
├──────────────────┬───────────────────────────────────────────────────────┤
│ Role             │ Primary Responsibilities & Scope                      │
├──────────────────┼───────────────────────────────────────────────────────┤
│ SuperAdmin       │ Platform owner. Full access to user management, course │
│                  │ approvals, and Permission Studio.                     │
│                  │ *Special rule*: Read-only oversight on support tickets. │
├──────────────────┼───────────────────────────────────────────────────────┤
│ SubAdmin         │ Delegated operational administrator. Handles course approvals │
│                  │ and works on support tickets assigned to them.        │
├──────────────────┼───────────────────────────────────────────────────────┤
│ ProductSupport   │ Helpdesk dispatcher & query solver. Exclusively triages │
│                  │ incoming tickets, resolves queries, and assigns tickets to │
│                  │ administrators. Cannot access general admin tools.    │
├──────────────────┼───────────────────────────────────────────────────────┤
│ Instructor       │ Content creator. Authors courses, defines syllabus sections, │
│                  │ uploads video lessons & PDFs, and creates quizzes.    │
├──────────────────┼───────────────────────────────────────────────────────┤
│ Student          │ Learner. Enrolls in courses, watches lectures, takes quizzes, │
│                  │ downloads certificates, and opens support tickets.    │
└──────────────────┴───────────────────────────────────────────────────────┘
```

Granular Support Desk Permissions Matrix

CAPABILITY	PRODUCT SUPPORT	SUBADMIN (ASSIGNED)	SUPERADMIN	STUDENT
View All Tickets	✅ Yes	✅ Queue / Assigned	✅ Yes (Read-Only)	❌ Only Own
Assign Ticket to Admin	✅ **Exclusive**	❌ Blocked (403)	❌ **Blocked (403)**	❌ Blocked
Change Ticket Status	✅ Yes	✅ Yes	❌ **Blocked (403)**	❌ Blocked
Post Staff Reply	✅ Yes	✅ Yes	❌ **Blocked (403)**	❌ Only Student Msg
Create Support Ticket	✅ Yes	✅ Yes	✅ Yes	✅ Yes

- **Backend Enforcement**: SupportController.cs
- **Frontend UI & Guards**: admin-support.ts, admin-support.html

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5. Course System, Video Player & PDF Resources

Course Lifecycle

```
[Instructor Creates Course] ──► [Draft State: IsPublished = false]
                                          │
                                          ▼ (Submits for Review)
                                [Admin / SubAdmin Reviews Course]
                                          │
                                          ▼ (Calls /api/Admin/courses/{id}/approve)
                                [Course Published: Visible in Marketplace]
```

Routing to the Learning Player

- When a student completes checkout or clicks "Continue Learning", Angular directs them to: `http://localhost:4201/learn/:courseId`
- **Route Guard & Backend Verification**:
- Frontend calls `GET /api/Learning/courses/{courseId}`.
- `LearningService.cs` verifies whether the student has an active record in `Enrollments`.
- **Privilege Override**: If the user is a `SuperAdmin`, `SubAdmin`, or the course author (`InstructorId == userId`), the backend unlocks **Preview Mode** so staff can inspect content without purchasing.
- Non-enrolled students are rejected with `401 Unauthorized` and redirected to the course landing page.

Player Capabilities

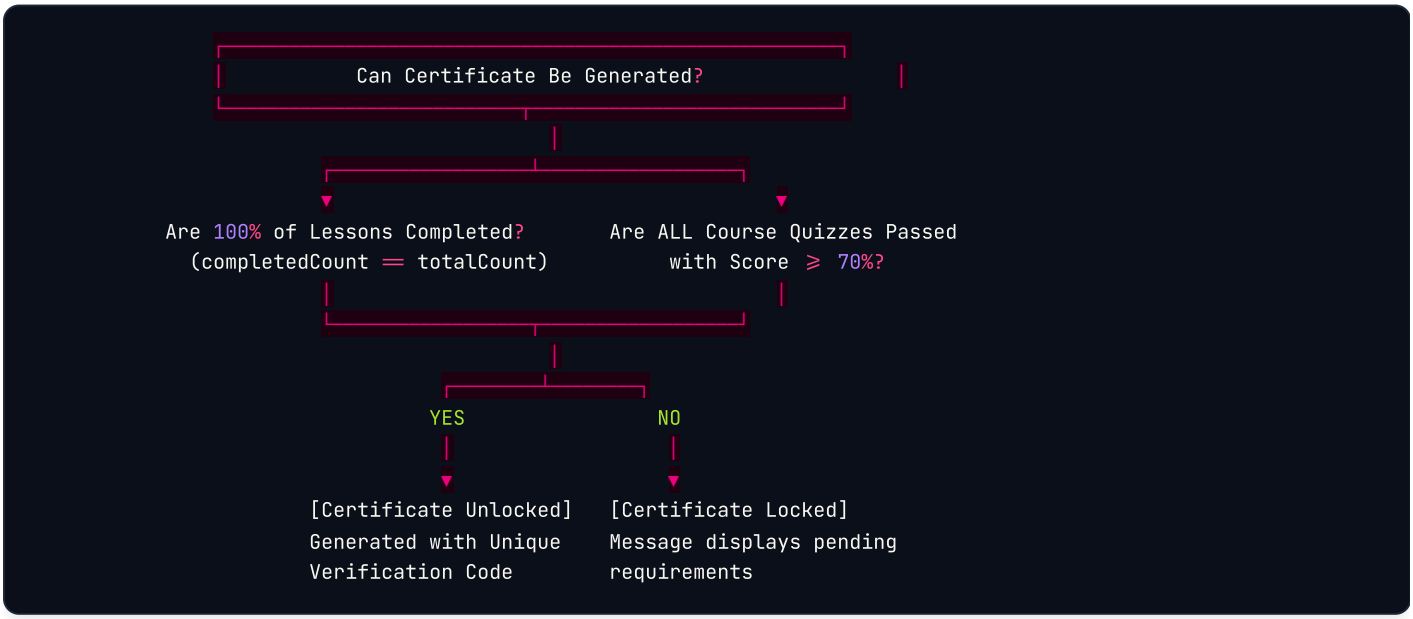
- **Video Playback**: Streams video URLs with automatic iframe or HTML5 video fallback.
- **Lesson Completion**: Toggle button updates `LessonCompletions` in real time, recalculates course completion percentage, and updates the circular progress bar.
- **Resource Downloads (PDFs)**:
- Lessons can have multiple attachments stored in `LessonResources`.
- Stored with `FileName` and `FileUrl`.
- Students can click resource badges to open reference PDFs and worksheets directly in a new tab or trigger downloads.
- **Key Files**:
- Backend: LearningController.cs, LearningService.cs
- Frontend: course-player.ts, course-player.html, course-player.scss

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6. Assessments, Quizzes & Strict 70% Certification Engine

The Strict 70% Rule

To ensure academic integrity, certificates cannot be unlocked simply by clicking through video lessons. SkillHub enforces a two-tier validation rule in `LearningService.cs`:



Grading Scale:

- **90% – 100%**: *Distinction*
- **80% – 89.9%**: *First Class with Distinction*
- **70% – 79.9%**: *Pass with Merit*
- **< 70%**: *Fail* (Must retake quiz to qualify for certificate)

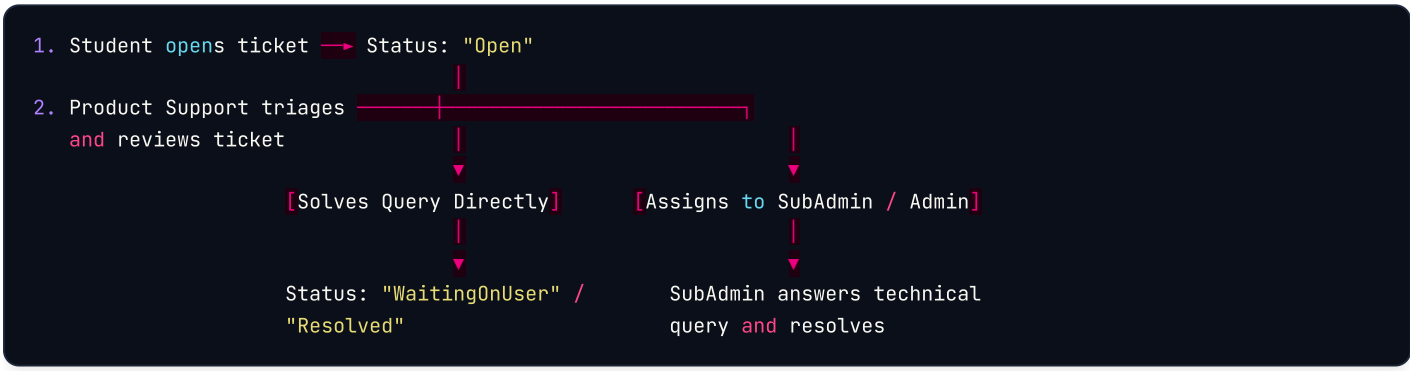
Certificate Attributes:

- Unique certificate code: `SKILLHUB-CERT-XXXXXXXX` (UUID-backed alphanumeric identifier).
- Dynamic printable view with student name, course title, completion date, verified instructor name, grade, and security seals.
- **Key Files**:
- Backend: LearningService.cs:L276–L380, QuizService.cs, Certificate.cs
- Frontend: course-player.ts:L350–L420, quiz.model.ts

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7. Helpdesk & Support Ticketing Workflow

Support Ticket Architecture & Life Cycle



Why SuperAdmin Has Read-Only Oversight:

In enterprise service desk operations, high-level executives and SuperAdmins maintain platform visibility to monitor SLA compliance, response times, and customer satisfaction metrics without interfering with day-to-day tier-1 and tier-2 operational queues.

- **Backend Enforcement**: In `SupportController.cs`:
- `UpdateTicketStatus`: If `User.IsInRole("SuperAdmin")` returns `403 Forbidden`.
- `AssignTicket`: If `!User.IsInRole("ProductSupport")` returns `403 Forbidden`.
- `AddMessage`: If `User.IsInRole("SuperAdmin")` and caller is not ticket owner returns `403 Forbidden`.
- **Assignable Admins Endpoint**: `GET /api/Support/admin/assignable-admins` fetches all active administrators (`SubAdmin` and `SuperAdmin`), enabling Product Support agents to route tickets dynamically.

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8. Payment Gateway & Atomic Enrollment

Payment Gateway Architecture

- Accessible at `/checkout/:courseId`.
- Completely removed all debug/appsettings notices to present a clean, high-trust production interface.

- Supported Payment Modes:
- **Credit / Debit Cards**: Includes a one-click **Test Card Fill** button (`4242 4242...`) for demonstrations.
- **UPI / QR Code**: Mock QR code display with support for Google Pay, PhonePe, Paytm, or manual VPA entry.
- **Net Banking**: Direct bank select chips for HDFC, ICICI, SBI, and Axis Bank.
- **Atomic Access Provisioning**: Once the transaction completes, the frontend sends an enrollment request to `POST /api/Courses/{courseId}/enroll` . An `Enrollment` record is committed, unlocking immediate access to the course player and redirecting to the congratulations screen.
- **Key Files**:
- payment-gateway.ts
- payment-gateway.html
- enrollment-success.html

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9. Google Gemini AI Copilot Integration

SkillHub integrates Google Gemini AI directly into the student experience via a floating, interactive copilot component.

```
[Student Browsing UI / Player]
            |
            ▼ (Sends prompt + current URL route)
[AiCopilotComponent (Angular)]
            |
            ▼ (HTTP POST /api/AiChat/message)
[AiChatController (ASP.NET Core)]
            |
            ▼
[GeminiAiService (Infrastructure)]
            |
            ▼ (REST call with API Key)
[Google Gemini Generative AI API (gemini-flash-latest)]
```

Key Technical Capabilities:

1. **Context-Aware Recommendations**:
2. The frontend passes `contextUrl: this.router.url` .
3. `GeminiAiService.cs` inspects the route (e.g. `/courses/5` vs `/learn/5` vs `/admin/support`) and tunes its prompt and quick suggestion chips to match what the student is looking at.
4. **Dual-Tier Quota Management**:
5. **Guests**: Rate-limited to 5 questions per session using an in-memory thread-safe `ConcurrentDictionary<string, int>` tracker keyed by IP address and guest session UUID.
6. **Authenticated Students**: Enjoy unlimited AI learning queries.
7. **Key Files**:
8. Backend: GeminiAiService.cs, AiChatController.cs
9. Frontend: ai-copilot.ts, ai-copilot.service.ts

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10. Advanced Technologies & Frameworks Explained

1. AutoMapper

- **Why We Use It**: Decouples database entities from public API responses.
- **Problem It Solves**: Without AutoMapper, developers must write dozens of manual mappings (`dto.Title = entity.Title; dto.InstructorName = entity.Instructor.FirstName + ...`). This is error-prone, repetitive, and risks exposing sensitive fields (such as `PasswordHash`).
- **Location**: MappingProfile.cs.

2. FluentValidation

- **Why We Use It**: Provides a clean, fluent syntax for building strongly typed validation rules without cluttering entity classes or DTOs with data annotations.
- **Rule Example**: `RuleFor(x ⇒ x.Email).NotEmpty().EmailAddress();`
- **Location**: `SkillHub.Application/Features/*/Validators/` .

3. BCrypt Password Hashing (`BCrypt.Net-Next`)

- **Why We Use It**: Employs adaptive salt-based hashing with configurable work factor (cost). Immune to rainbow table attacks and brute force cracking.
- **Location**: UserSeeder.cs, AuthService.cs.

4. Angular 20 Standalone Components & Signals

- Eliminates legacy `NgModule` boilerplate.
- Promotes modularity, faster tree-shaking, smaller bundle sizes, and clean lazy loading.
- Functional HTTP interceptor (`jwtInterceptor`) automatically injects Bearer authorization headers and intercepts 401s.

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11. Complete Codebase File Index

MODULE / TOPIC	BACKEND FILE (ASP.NET CORE)	FRONTEND FILE (ANGULAR)
Authentication & JWT	`SkillHub.API/Controllers/AuthController.cs` `SkillHub.Infrastructure/Services/Identity/AuthService.cs`	`src/app/features/auth/login/login.ts` `src/app/core/interceptors/jwt.interceptor.ts`
Clean Architecture Root	`SkillHub.Domain/Entities/` `SkillHub.Application/Mappings/MappingProfile.cs`	`src/app/app.routes.ts`
Database & Context	`SkillHub.Infrastructure/Persistence/Context/ApplicationDbContext.cs`	—
Seed Data (Users/Roles)	`SkillHub.Infrastructure/seed/UserSeeder.cs` `SkillHub.Infrastructure/seed/RoleSeeder.cs`	—

MODULE / TOPIC	BACKEND FILE (ASP.NET CORE)	FRONTEND FILE (ANGULAR)
Permission Studio	`SkillHub.API/Controllers/PermissionController.cs` `SkillHub.Infrastructure/Services/Admin/PermissionService.cs`	`src/app/features/admin/admin-permissions/admin-permissions.ts` `src/app/features/admin/services/permission.service.ts`
Course Authoring	`SkillHub.API/Controllers/CourseController.cs` `SkillHub.Infrastructure/Services/Courses/CourseService.cs`	`src/app/features/courses/course-create/course-create.ts` `src/app/features/courses/course-list/course-list.ts`
Course Player & PDFs	`SkillHub.API/Controllers/LearningController.cs` `SkillHub.Infrastructure/Services/Learning/LearningService.cs`	`src/app/features/enrollment/course-player/course-player.ts` `src/app/features/enrollment/course-player/course-player.html`
Quizzes & 70% Certificate	`SkillHub.API/Controllers/QuizzesController.cs` `SkillHub.Infrastructure/Services/Assessments/QuizService.cs` `SkillHub.Domain/Entities/Learning/Certificate.cs`	`src/app/features/enrollment/course-player/course-player.ts` `src/app/features/enrollment/models/quiz.model.ts`
Helpdesk & Support	`SkillHub.API/Controllers/SupportController.cs` `SkillHub.Infrastructure/Services/Support/SupportService.cs`	`src/app/features/admin/admin-support/admin-support.ts` `src/app/features/support/services/support.service.ts`
Payment Gateway	`SkillHub.API/Controllers/PaymentsController.cs` `SkillHub.Infrastructure/Services/Payments/PaymentService.cs`	`src/app/features/enrollment/payment-gateway/payment-gateway.ts` `src/app/features/enrollment/payment-gateway/payment-gateway.html`
Google Gemini AI	`SkillHub.API/Controllers/AiChatController.cs` `SkillHub.Infrastructure/Services/Ai/GeminiAiService.cs`	`src/app/features/ai-copilot/ai-copilot.ts` `src/app/features/ai-copilot/services/ai-copilot.service.ts`

12. Viva / Practical Exam Top Questions & Answers

Q1: What is Clean Architecture and why did you choose it over N-Tier architecture?

> **Answer**: Clean Architecture organizes software into concentric layers where dependencies strictly point inward towards the core Domain (`SkillHub.Domain`). In traditional N-Tier architectures, business logic often becomes tightly coupled to Entity Framework or database tables. In Clean Architecture, our business rules (`Application` layer) and domain models (`Domain` layer) are completely independent of UI technologies, third-party libraries, and database providers.

Q2: How does Angular communicate securely with ASP.NET Core?

> **Answer**: Angular communicates with the Web API using JSON over HTTPS. When a user logs in, the backend issues a signed **JWT (JSON Web Token)** containing identity claims (`userId` , `email` , `role`). Angular stores this token securely and utilizes a functional **HTTP Interceptor** (`jwt.interceptor.ts`) that automatically attaches the `Authorization: Bearer <token>` header to all outgoing requests. The backend validates the signature, issuer, and expiration on every request.

Q3: Explain how your 70% certificate rule is enforced in code.

> **Answer**: In `SkillHub.Infrastructure.Services.Learning.LearningService.cs` , when a lesson is completed, the system calls `EvaluateCertificateEligibilityAsync` . It executes two checks: 1. Counts completed lessons against total section lessons (`completedLessons ≥ totalLessons`). 2. Queries `QuizAttempts` for the student on that course and verifies that for every quiz, `bestAttempt.ScorePercentage ≥ 70` . Only when both conditions evaluate to `true` does the backend generate a unique `Certificate` record with a verifiable UUID code (`SKILLHUB-CERT-XXXXXXXX`).

Q4: Why is SuperAdmin restricted to read-only oversight on support tickets?

> **Answer**: In real-world enterprise architectures, ticket handling is an operational responsibility delegated to **Product Support** and assigned admins (`SubAdmin`). The **SuperAdmin** role represents platform ownership and auditing. Giving SuperAdmin read-only oversight ensures separation of duties: SuperAdmin can monitor platform SLAs and customer issues without accidentally tampering with active ticket triage or assignments. This is enforced with a `403 Forbidden` guard in `SupportController.cs` .

Q5: What is the purpose of AutoMapper and FluentValidation in your project?

> **Answer**: - **AutoMapper** prevents boilerplate code and eliminates "over-posting" vulnerabilities by transforming database entities into lightweight DTOs before sending them to the client. - **FluentValidation** decouples validation logic from model definitions. Instead of using messy attributes, we write expressive, testable validation rules in dedicated validator classes that execute before controller actions run.

Q6: How does the Google Gemini AI integration handle guest usage versus student usage?

> **Answer**: `GeminiAiService.cs` tracks unauthenticated guest sessions in a thread-safe `ConcurrentDictionary` by IP and guest session ID, capping them at 5 exploratory queries to prevent API abuse. Authenticated students are granted unlimited queries and receive contextual suggestions tuned to their active route.

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13. System Demo User Accounts

ROLE	EMAIL	PASSWORD	CAPABILITIES TO DEMONSTRATE
SuperAdmin	`superadmin@skillhub.com`	`SuperAdmin123!`	User management, Permission Studio, Course Approvals, Read-Only Support Oversight
ProductSupport	`support@skillhub.com`	`Support123!`	Helpdesk Command: Triage tickets, assign to SubAdmins, resolve tickets
SubAdmin	`subadmin@skillhub.com`	`SubAdmin123!`	Operational Admin: Resolve assigned tickets, course management
Instructor	`instructor@skillhub.com`	`Instructor123!`	Authoring courses, managing sections, uploading lesson videos and PDFs
Student	`student@skillhub.com`	`Student123!`	Enrolling, course player, taking quizzes, earning certificates, raising tickets

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Created for SkillHub Project Presentation, Practical Examination & Viva.